PHASE CONTROLLED RECTIFIERS

Module 2

POWER CONVERTERS

• Power Electronic converter converts the available form of voltage into a form with different magnitude or frequency.

TYPES:-

- PHASE CONTROLLED RECTIFIERS (AC to DC)
- CHOPPERS (DC to DC)
- INVERTERS (DC to AC)
- AC VOLTAGE CONTROLLERS or AC REGULATORS (fixed AC to variable AC)
- CYCLOCONVERTERS (AC to AC with different frequency)

POWER CONVERTERS

- **PHASE CONTROLLED RECTIFIERS**

  - converts constant AC input to controllable DC output

  - uses line voltage for commutation, so it is called line commutated or naturally commutated AC to DC converters.

  - may be 1-phase or 3-phase

  - used in DC drives, metallurgical and chemical industries, as excitation systems for synchronous machines.

- **CHOPPERS**

  - Converts fixed DC input to variable DC output

  - Requires forced commutation to turn off SCR

  - classified depending on the type of commutation and the direction of power flow

  - used in DC drives, trolley trucks, battery driven vehicles etc.

POWER CONVERTERS

- **INVERTERS**

  - converts fixed DC voltage to variable AC voltage

  - output may be a variable voltage and variable frequency

  - use natural or forced commutation for turning off the thyristors

  - used in induction motor and synchronous motor drives, induction heating, UPS, HVDC transmission systems etc.

POWER CONVERTERS

- **AC VOLTAGE CONTROLLERS (or AC REGULATORS)**

  - converts fixed AC voltage to variable AC voltage at same frequency

  - natural commutation is used for turn off of the thyristor

  - used in lighting control, speed control of fans, pumps etc

- **CYCLOCONVERTERS**

  - converts input power at one frequency to output power at different frequency

  - Line commutation is commonly used but forced commutation is also used

  - used for slow-speed large AC drives like rotary kilns

PHASE CONTROLLED CONVERTERS (Contd…)

- Rectification – process of converting an alternating voltage or current into a direct voltage or current and can be achieved by a variety of circuits
- Widely used switching devices- diodes, power transistors, power MOSFETs, thyristors etc.
- Converts constant AC input voltage to controllable DC output voltage
- Thyristor turns off naturally when the supply voltage reverses and the anode current must be less than holding current.
- Thus natural commutation occurs in phase controlled rectifiers.
- Natural commutation – turning off of the thyristor by the supply voltage itself.
- Forced commutation - turning off of a thyristor forcibly by using passive components.

PHASE CONTROLLED CONVERTERS (Contd…)

- When more than one SCR is used as in industrial applications, when the incoming SCR is turned on, it immediately reverse biases the outgoing SCR and thus it is turned off.
- Rectifier circuits – 3 types- uncontrolled, fully-controlled & half-controlled.
- **Uncontrolled rectifier** – uses only diodes, amplitude of the DC output voltage is fixed
- **Fully-controlled rectifier -** uses only thyristors as rectifying elements, amplitude of the DC output voltage is a function of the ac supply voltage and the point on the wave at which the thyristors are triggered.
- **Half-controlled rectifier** – mixture of diodes & thyristors, allowing a limited control over the DC output voltage, cheaper than fully-controlled rectifier of same rating, has operational limitations.

PHASE CONTROLLED RECTIFIERS

Controllable DC power – industrial applicatons are :-

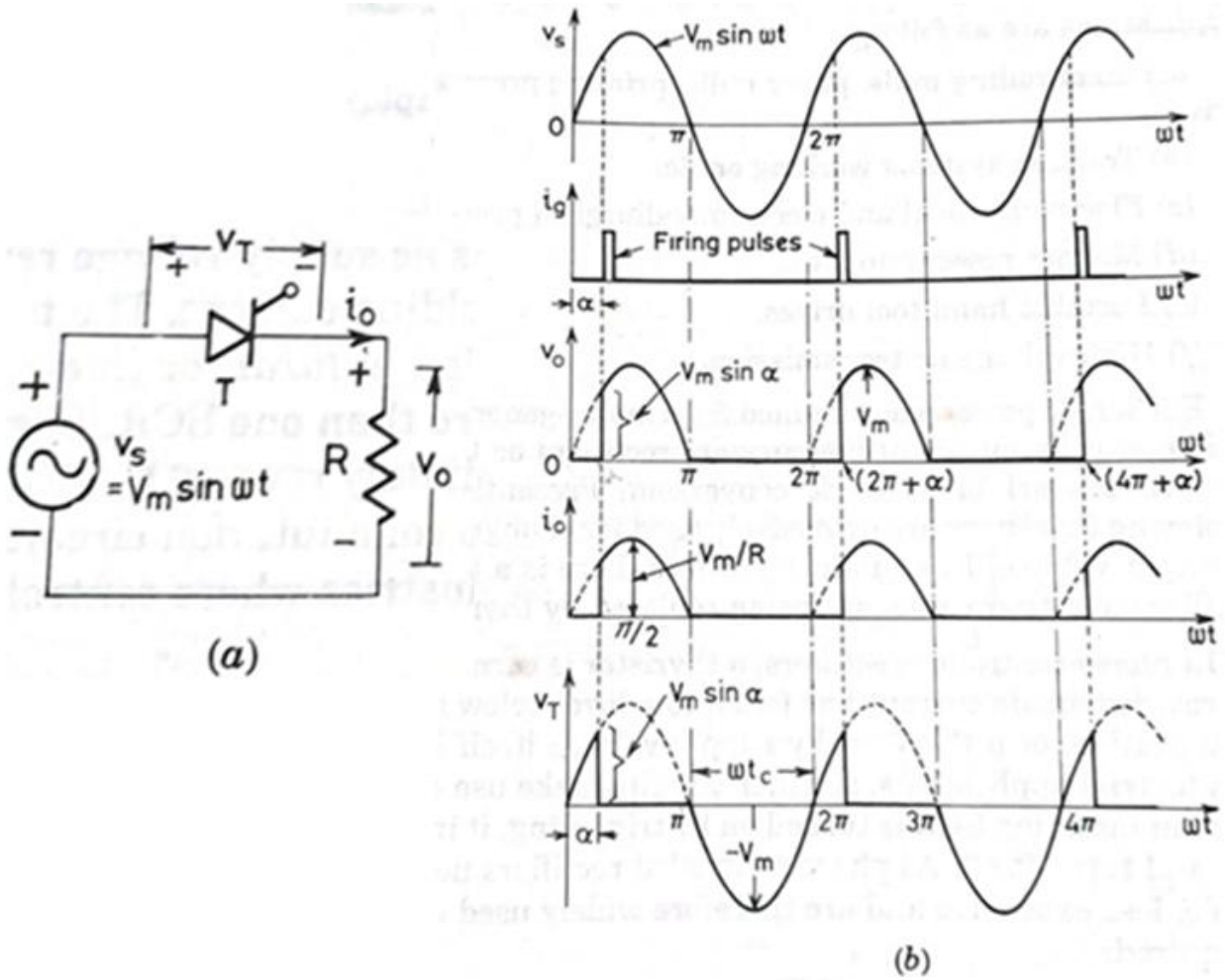
- Steel rolling mills, paper mills, printing press, textile mills employing DC motor drives
- Traction systems working on DC
- Electrochemical & metallurgical processes
- Portable hand tool drives
- HVDC transmission

CONTROLLED RECTIFIERS

HALF WAVE – R, RL, RLE LOAD

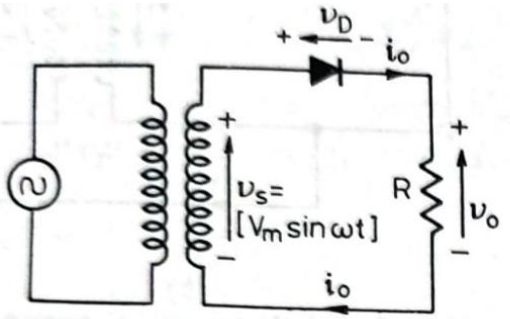
Module 2

Single phase half wave thyristor circuit- R load

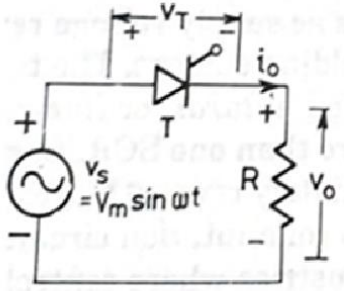


Single-phase half-wave thyristor circuit with R load
(a) circuit diagram and (b) voltage and current waveforms.

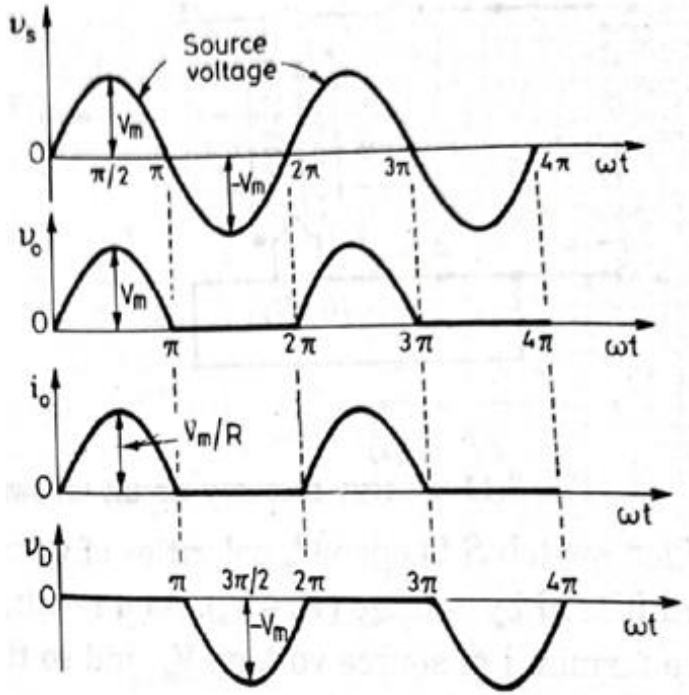
Half-wave Rectifier (Uncontrolled & Controlled)- Comparison

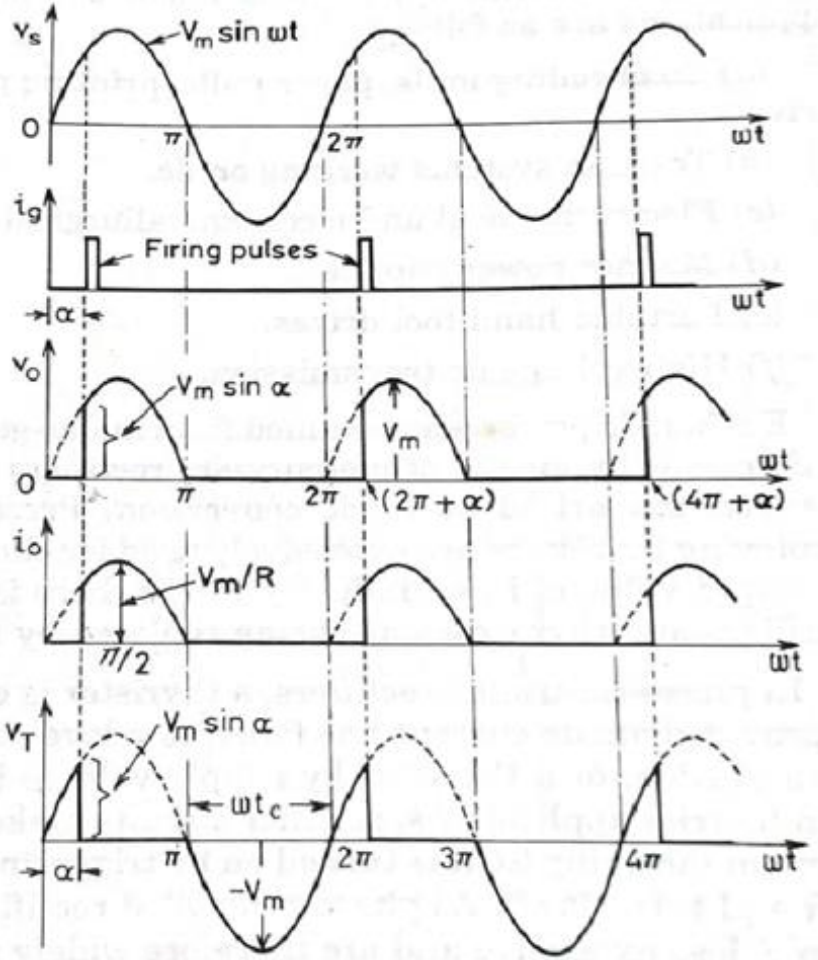


Single-phase half-wave diode rectifier with R load



Single-phase half-wave thyristor circuit with R load





Single phase half wave thyristor circuit- R load

- Average voltage V_o across load R in terms of the firing angle α,

$$V_o = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin\omega t.d(\omega t) = \frac{V_m}{2\pi}(1+\cos\alpha)$$

- The maximum voltage occurs at α=0°,

$$V_{om} = \frac{V_m}{2\pi}(1+\cos 0) = \frac{V_m}{\pi}$$

- Average load current ,

$$I_o = \frac{V_o}{R} = \frac{V_m}{2\pi R}(1+\cos\alpha)$$

Single phase half wave thyristor circuit- R load

To find RMS value of load voltage, V_{or}:-

- RMS value of load voltage, V_{or}

$$V_{or} = \left[\frac{1}{2\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t . d(\omega t) \right]^{1/2} = \frac{V_m}{2\sqrt{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

- RMS value of load current, I_{or}

$$I_{or} = \frac{V_{or}}{R}$$

- Power delivered to the load= (rms load voltage)(rms load current)

$$= V_{or} I_{or} = \frac{V_{or}^2}{R} = I_{or}^2 R$$

Single phase half wave thyristor circuit- R load

- Input volt-ampere(volt-ampere rating of transformer)

=(rms source voltage)(total rms line current)

$$= V_s I_{or} = \frac{\sqrt{2}V_s^2}{2R\sqrt{\pi}}\left[(\pi-\alpha)+\frac{1}{2}\sin 2\alpha\right]^{1/2}$$

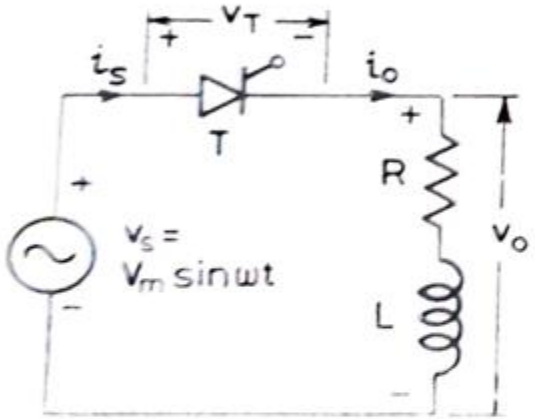
- Input power factor

$$= \frac{Power\ delivered\ to\ load}{InputVA} = \frac{V_{or}.I_{or}}{V_s.I_{or}} = \frac{V_{or}}{V_s}$$

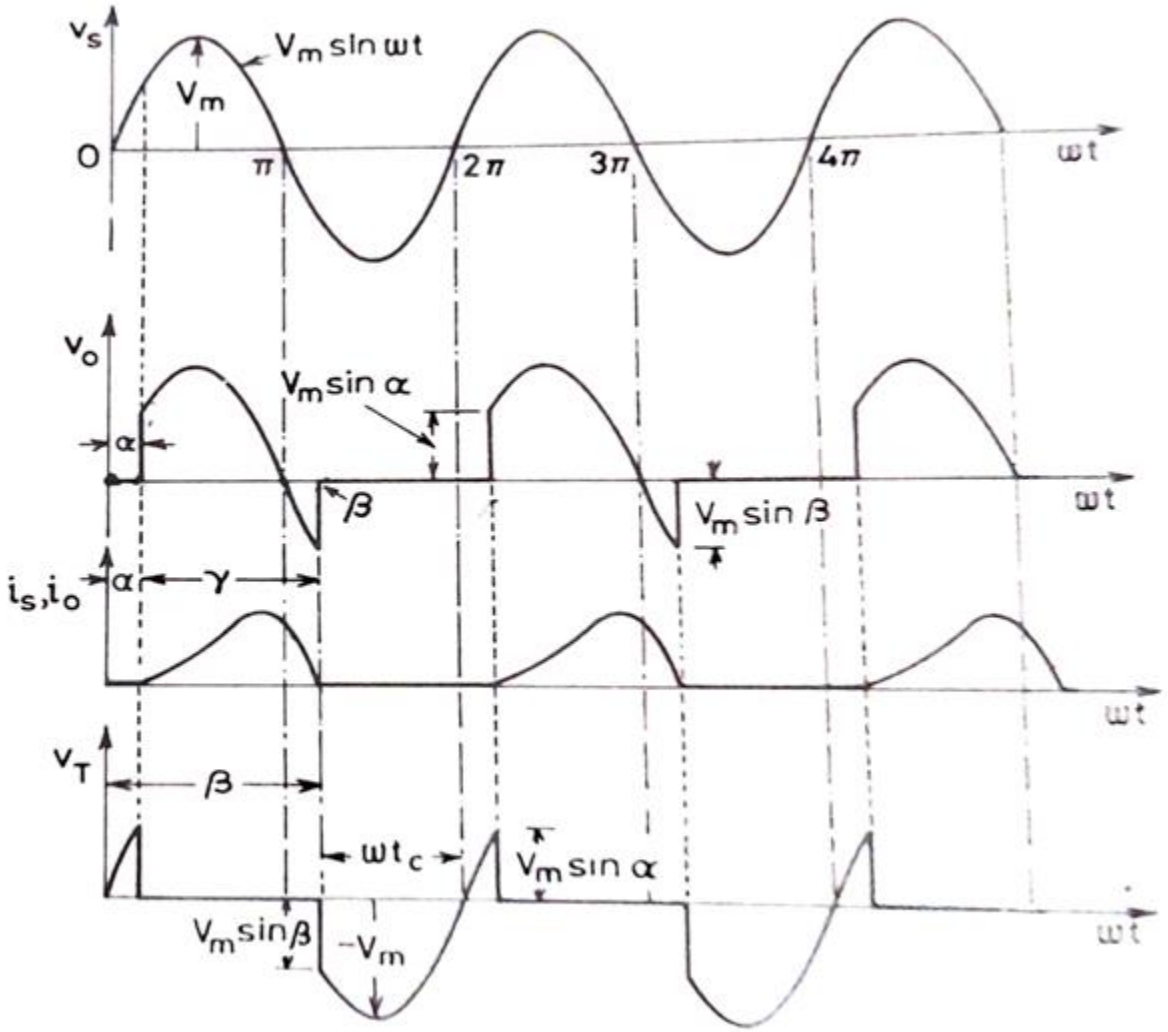
- Input pf

$$= \frac{1}{\sqrt{2}\pi}\left[(\pi-\alpha)+\frac{1}{2}\sin 2\alpha\right]^{1/2}$$

Single phase half wave thyristor circuit- RL load



(a)



(b)

Single phase half wave thyristor circuit- RL load

- **At ωt=α**, the thyristor T is turned on. Immediately load voltage v_o becomes equal to source voltage v_s. Because of the load inductance L the load current i_o rise gradually from zero at α and after some time it reaches a maximum value and then begin to decrease.
- **At ωt=π**, $v_o = 0$, but i_o is not zero because of inductance L.
- **After ωt=π**, thyristor T is subjected reverse anode voltage, but will not turn-off as the anode current of the SCR which is the load current is not less than the holding current.
- **At some angle β > π**, i_o reduces to zero and SCR turns off as it is already biased.
- **After ωt=β,** $v_o = 0$ & $i_o = 0$**.**
- **At ωt=2π+α,** SCR is triggered again, v_o is applied to the load & the load current also flows as before.

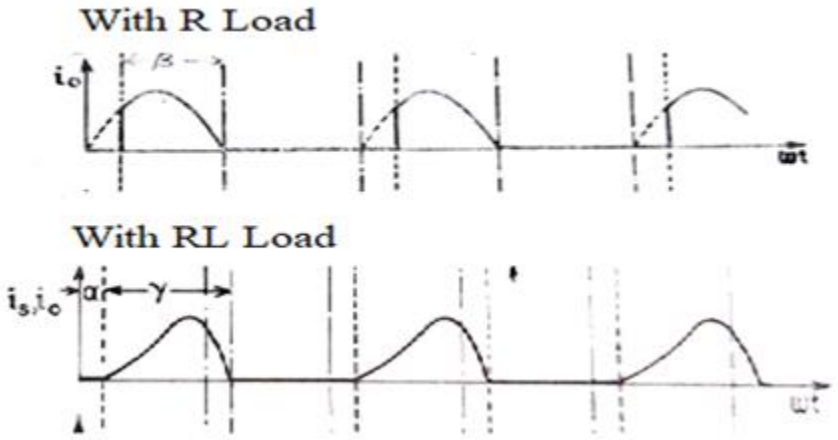
Single phase half wave thyristor circuit- RL load

- Angle β – is called extinction angle
- Angle $(\beta-\alpha)= \gamma$ is called conduction angle
- At $\omega t=\alpha$, $v_T = V_m \sin\alpha$
- From $\omega t=\alpha$ to β, $v_T = 0$
- At $\omega t=\beta$, $v_T = V_m \sin\beta$
- As $\beta > \pi$, v_T is negative at $\omega t=\beta$. So the thyristor is reverse biased from $\omega t=\beta$ to 2π.
- Circuit turn-off time $t_c = (2\pi - \beta)/\omega$ seconds
- For satisfactory commutation t_c should be greater than SCR turn-off time t_q.

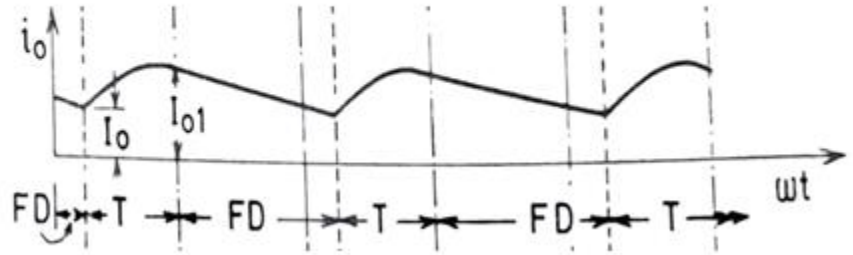
Single phase half wave thyristor circuit- RL load & FWD

- This circuit improves the output current waveform with the help of a *freewheeling diode(FWD).*
- Also known as commutating diode or by-pass diode

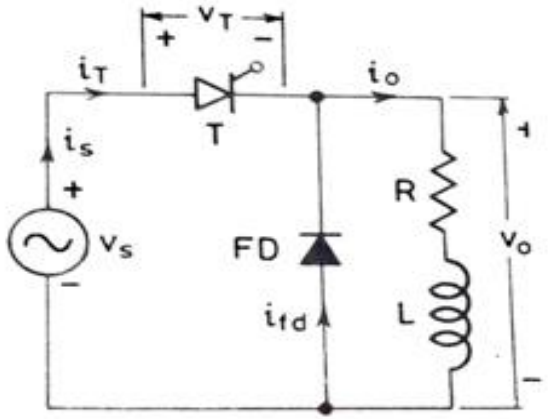
- Discontinuous current mode



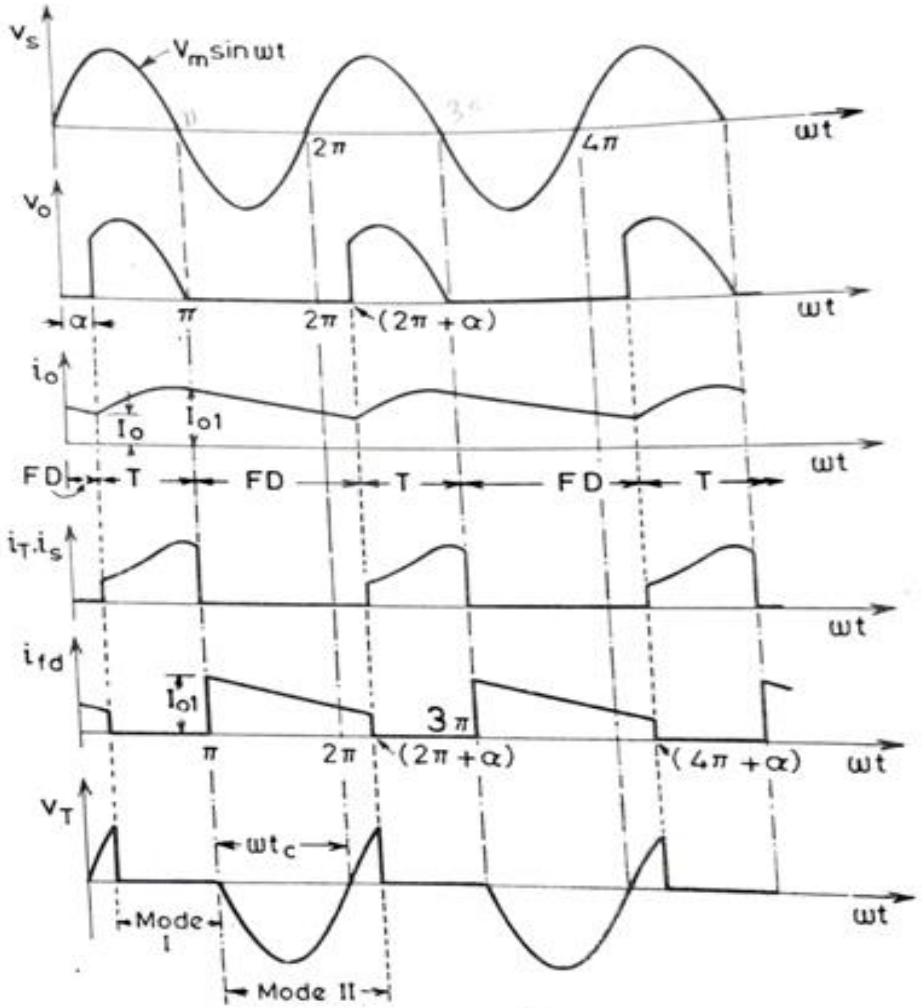
- Continuous current mode



Single phase half wave thyristor circuit- RL load & FWD



(a)



(b)

Single phase half wave thyristor circuit- RL load & FWD

- At $\omega t = 0$, v_s is becoming +ve, SCR is forward biased.
- At $\omega t = \alpha$, SCR is triggered & $v_s = v_o$.
- At $\omega t = \pi$, $v_s = 0$
- Just after $\omega t = \pi$, v_s tends to reverse, freewheeling diode FD is forward biased through the conducting SCR. (SCR is still conducting because the current through it is not less than the holding current). Now i_o is immediately transferred from SCR to FD. The SCR is now subjected to reverse voltage and zero current. Therefore SCR is turned off.
- During freewheeling period i_o does not decay to zero until it is triggered again at $\omega t = 2\pi + \alpha$.
- The voltage across the freewheeling diode during conduction is taken as zero. Therefore the load voltage is zero during this period.

Single phase half wave thyristor circuit- RL load & FWD

- SCR is reverse biased from π to 2π.
- Therefore, circuit turn-off time, t_c = time during which SCR is reverse biased/ω =$(2\pi-\pi)/\omega = \pi/\omega$ seconds, where $\omega= 2\pi f$, f=supply frequency in Hz.
- The SCR is connected in series with the supply. Therefore the supply current & the SCR have the same waveshape.

Single phase half wave thyristor circuit- RL load & FWD

- 2 modes of operation for this circuit.
  - Mode 1- Conduction mode
  - Mode 2 – Freewheeling mode

  - Mode 1- Conduction mode
    - SCR conducts from α to π, $2\pi+\alpha$ and so on
    - Freewheeling diode is reverse biased
    - Duration of this mode = $(\pi-\alpha)/\omega$ seconds
  - Mode 2 – Freewheeling mode
    - Load current is assumed continuous.
    - Freewheeling diode conducts from π to $2\pi+\alpha$, 3π to $4\pi+\alpha$ and so on
    - SCR is reverse biased

Single phase half wave thyristor circuit- RL load & FWD

- Load current is contributed by
  - SCR from α to π, $(2\pi+ \alpha)$ to 3π and so on
  - FD from 0 to α to π, $(2\pi+ \alpha)$ to 3π and so on
- SCR current waveshape & load current waveshape is identical for α to π, $(2\pi+ \alpha)$ to 3π and so on.
- FD current waveshape & load current waveshape is identical for 0 to α to π, $(2\pi+ \alpha)$ to 3π and so on.
- Load absorbs power(p_1)from source from α to π(both v_o & i_o +ve).
- The energy stored in L(p_2) is returned to the source from π to $(2\pi+ \alpha)$ (v_o is –ve & i_o is +ve)
- Net power consumed by load is the difference between p_1 & p_2

Single phase half wave thyristor circuit- RL load & FWD

- Load absorbs power from α to π but for π to $(2\pi+ \alpha)$, the energy stord in L is delivered to R through FD. Thus the power consumed by the load is more.
- i.e., the power delivered to the load, for the same firing angle is more when freewheeling diode is used.
- Input powerfactor(=power delivered to load /input volt-ampere) is improved with the use of freewheeling diode.
- Load current waveform is also improved with the use of freewheeling diode.

Single phase half wave thyristor circuit- RL load & FWD

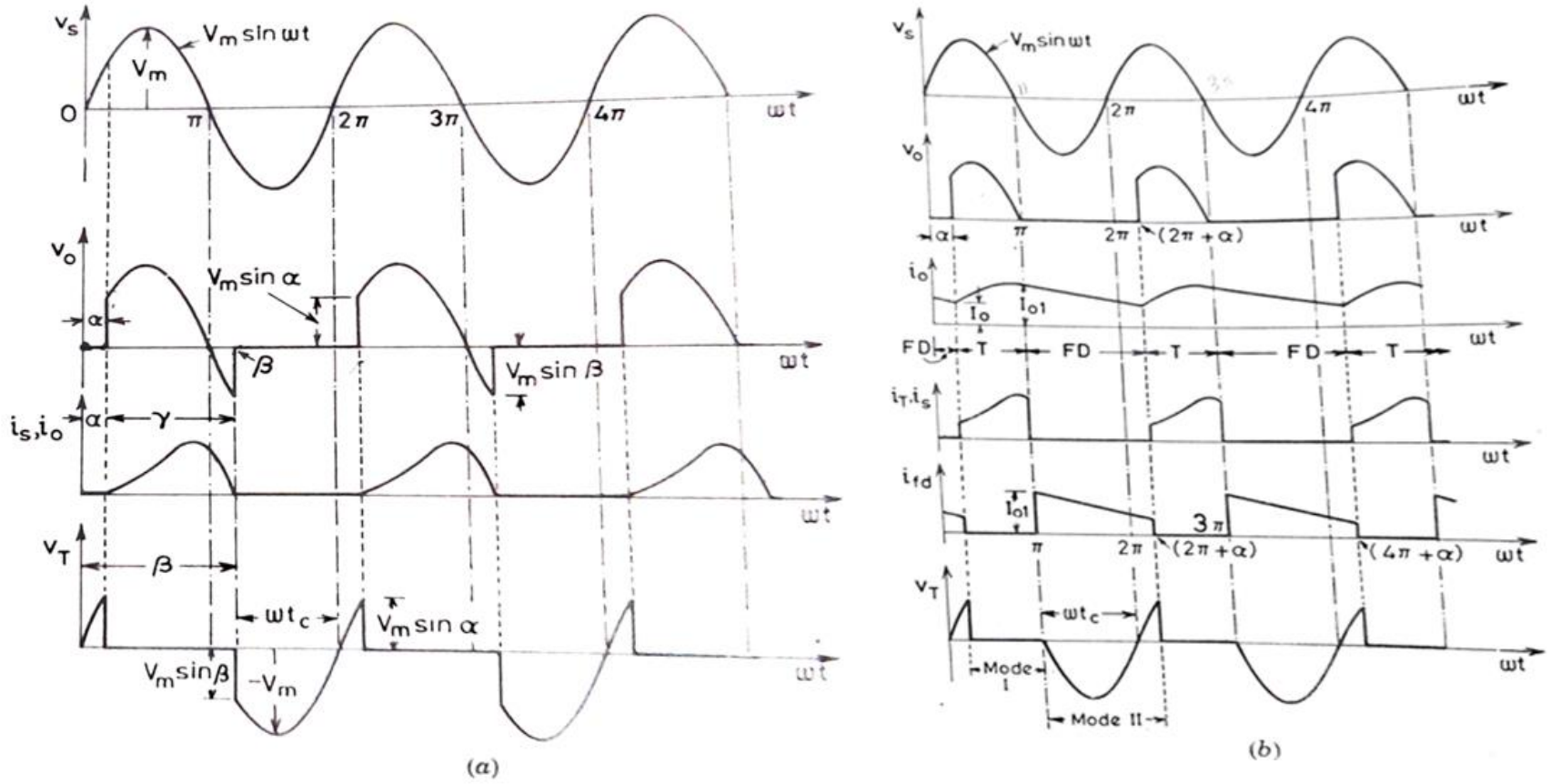
Advantages of using freewheeling diode:-

- Input powerfactor is improved.
- Load current waveform is improved.
- Load performance is better.
- Prevents the load voltage from becoming negative.
- When the load voltage tends to become negative, the load current is transferred from thyristor to freewheeling diode.

Single phase half wave thyristor circuit- RL load & FWD

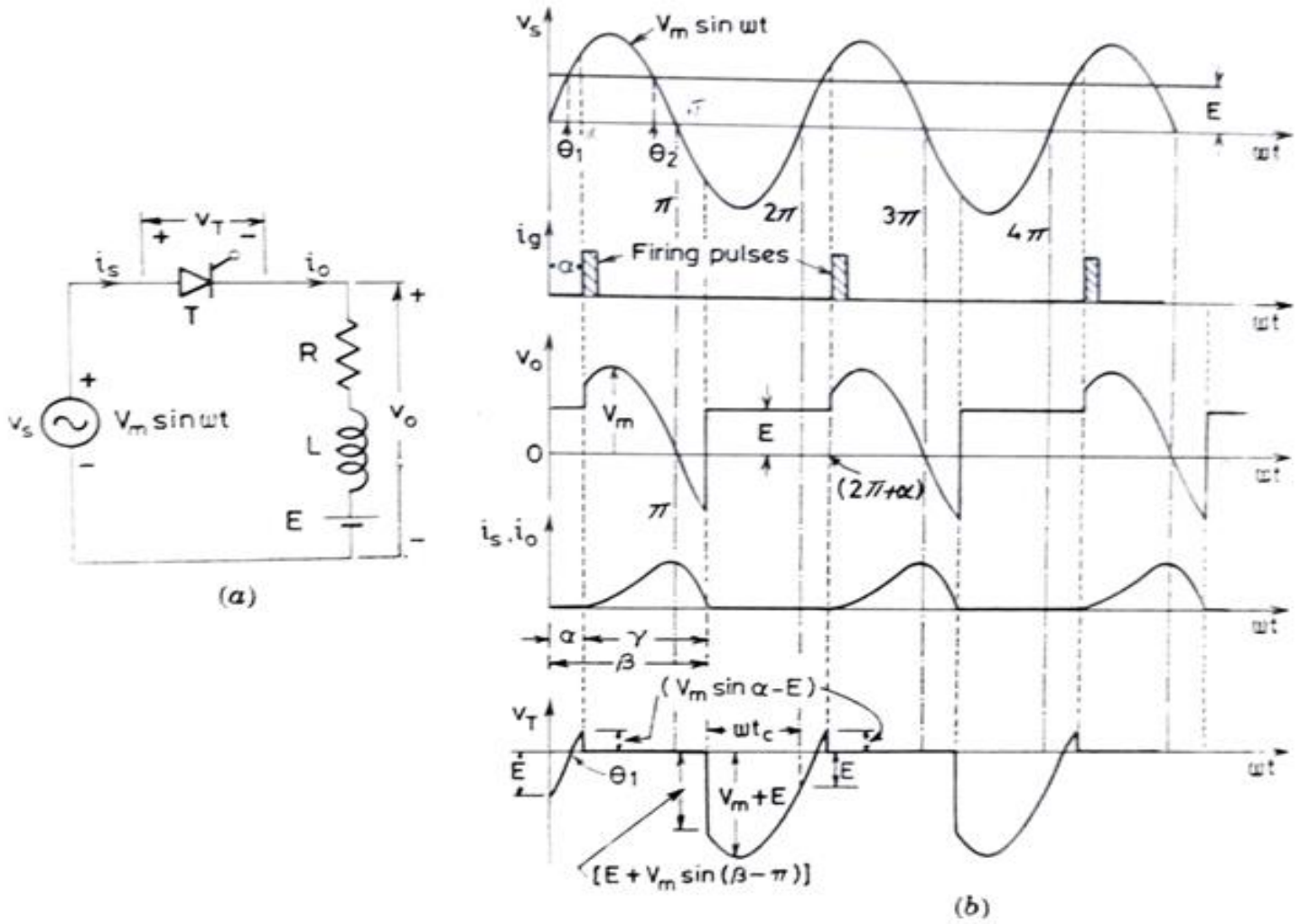
- Supply current i_s is unidirectional and is in the form of DC pulses.
- Thus single phase half wave converter introduces a DC component into the supply line.
- This is not desirable, as it leads to saturation of the supply transformer and also introduces harmonics.
- These disadvantages can be overcome by using single phase full wave circuits

Single phase half wave thyristor circuit



Waveforms for half-wave controlled rectifier
(a) without FWD (b) with FWD

Single phase half wave thyristor circuit- RLE load



Single phase half wave thyristor circuit- RLE load

- The load comprises a series combination of R & L in series with a back emf E.
- The emf E in the load may be a battery or a DC motor.
- The firing angle(α) is limited between the angle θ_1 and θ_2.(refer waveform)
- The firing angle(α) cannot be less than θ_1 and more than θ_2 because the anode of SCR is more negative than the cathode.
- Minimum value of firing angle is obtained from $V_m \sin\omega t = E$. this occurs at where $\theta_1 = \sin^{-1}(E/V_m)$
- If thyristor is triggered at an angle $\alpha < \theta_1$, the $E > V_s$, SCR is reverse biased and it will not turn on.
- Maximum value of firing angle is $\theta_2 = \pi - \theta_1$.
- When i_o is zero, v_o = E because there is no volage drop in R & L.
- When i_o is not zero, $v_o = v_s$